

Cover Letter

Student name: Abu Hanif

Course: DSA 502

Project: Movie Metadata RAG Assistant (Hanif_Movie_metadata)

Dear Instructor,

Please accept this submission for my mini-project Movie Metadata RAG Assistant. The application is a Flask web service that loads movie metadata from `movies_metadata.csv`, cleans and normalizes key fields (including human-readable genres), and implements retrieval with TF-IDF vectorization and cosine similarity to surface the top matching films for each user question. Retrieved rows are packaged into a concise context block and sent to the Ollama HTTP API so an LLM can produce a grounded response based strictly on that context.

This work demonstrates practical integration of classical information retrieval, a Python data pipeline, and modern LLM response generation in a single end-to-end workflow suitable for coursework in data structures, algorithms, and applied systems.

Screenshots (app in the browser)

The following images are running-app captures from this project (files live in `screenshots/` in the repository).

Home - search bar and example prompts

Movie Metadata RAG Assistant - home screen

Movie Metadata RAG Assistant

TF-IDF retrieval over TMDB-style metadata · grounded answers via Ollama

Ask about genres, themes, eras, or titles...

Search & Ask

EXAMPLE QUESTIONS

Animated family comedy about toys

Romance and drama in New York City

Space science fiction adventure

Psychological thriller mystery

Magical board game adventure

Flask · scikit-learn TF-IDF + cosine similarity · Ollama `minimax-m2.1:cloud`

Example: "Best comedy movie" - retrieval table

Movie Metadata RAG Assistant - best comedy search

Movie Metadata RAG Assistant

TF-IDF retrieval over TMDB-style metadata · grounded answers via Ollama

Best comedy movie

Search & Ask

EXAMPLE QUESTIONS

Animated family comedy about toys

Romance and drama in New York City

Space science fiction adventure

Psychological thriller mystery

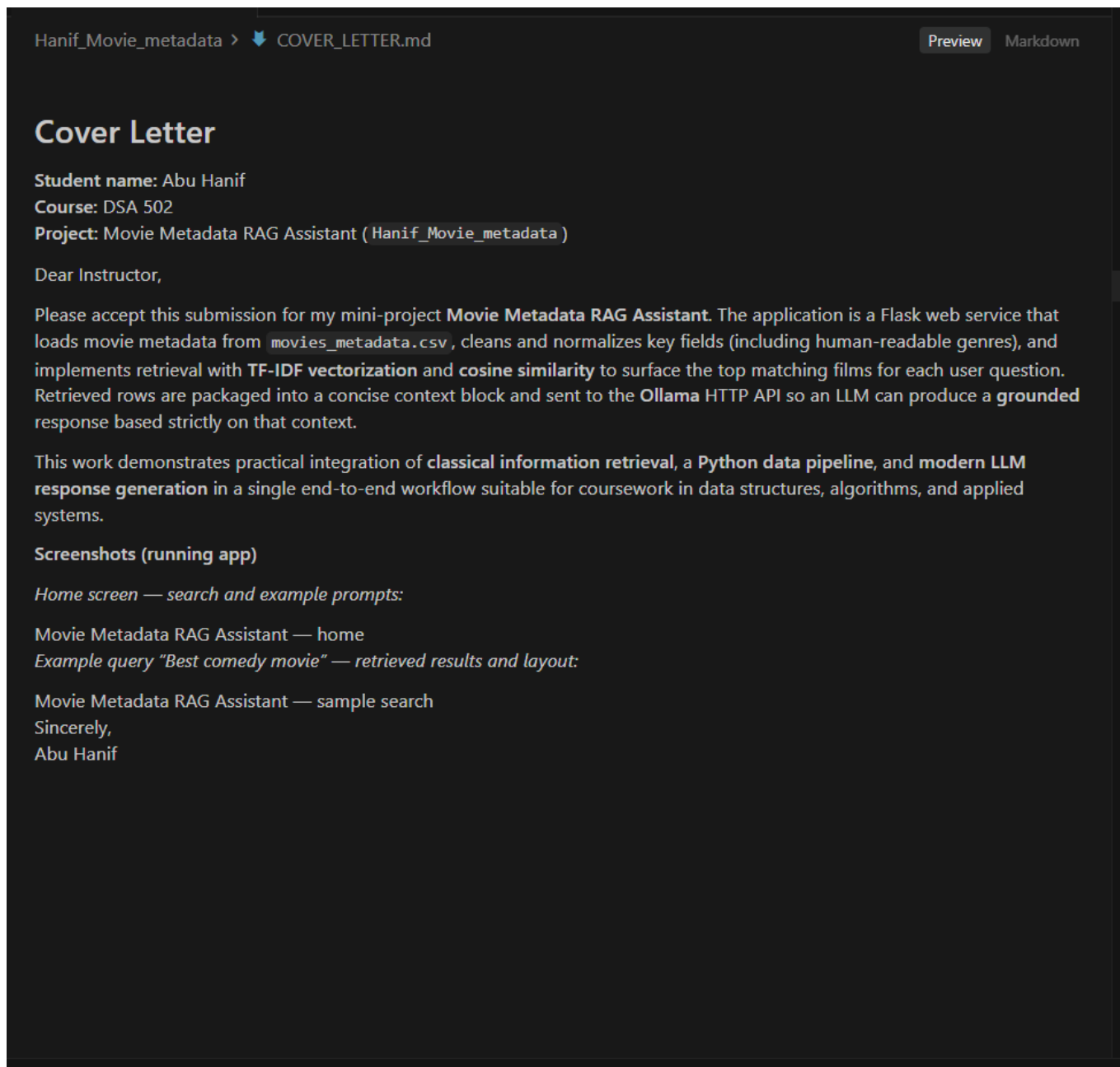
Magical board game adventure

Retrieved movies (top 5)

RANK	TITLE	GENRES	RELEASE DATE	VOTE AVG	VOTE COUNT	SIMILARITY SCORE	OVERVIEW
1	The Best Movie 3-DE	Comedy	2011-01-19	3.0	8	0.5087	A third part of "The Best Movie" trilogy.
2	The Best Movie 2	Comedy	2009-01-22	3.0	9	0.3576	The Best Movie 2 is a 2009 Russian comedy film from Monumental Pictures (Sony Pictures Entertainment) and Comedy Club Production, continuation of 2008 The Best Movie film, spoofing such famous Russian films as Zhara, The Irony of Fate 2 and TV-programs: Malakhov+, King of Ring and Taxi.
3	Disaster Movie	Action, Comedy	2008-08-29	3.1	250	0.3008	In DISASTER MOVIE, the filmmaking team behind the hits "Scary Movie," "Date Movie," "Epic Movie" and "Meet The Spartans" this time puts its unique, inimitable stamp on one of the biggest and most bloated movie genres of all time -- the disaster film.
4	Movie Movie	Comedy	1978-11-01	4.5	4	0.2627	Three movie genres of the 1930s, boxing films, WWI aviation dramas, and backstage Broadway musicals, are satirized using the same cast.
5	Promise	TV Movie, Drama, Family	1986-12-21	0.0	0	0.2489	Promise, the most honored movie in television history, is a deeply moving story of family love and loyalty. Bob (James Garner) returns home for his mother's funeral knowing that he must fulfill a promise made 30 years earlier—to care for his younger brother D.J. (James Woods), who is schizophrenic. When Bob eventually takes over responsibility for his younger brother's well-being, he embarks upon the most difficult challenge of his life. Following its premiere in 1986, Promise won virtually every award a television program can win, including five Emmys® (Best Actor, Best Special, Best Direction, Best Teleplay, Best Writing), two Golden Globes®, a Peabody Award, a Christopher Award, the Humanitas Prize and several other honors. The movie remains an enduring favorite in the Hallmark Hall of Fame's library of films.

Example: "Romance and drama in New York City" - retrieval table

Movie Metadata RAG Assistant - romance and drama NYC



Example: "Movies similar to a magical board game adventure" - retrieval table

Movie Metadata RAG Assistant - magical board game query



Example: additional search - retrieval table

Movie Metadata RAG Assistant - additional sample search



Sincerely,
Abu Hanif